3DWEBSOFT INC. FOUNDATION

FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Class 39: Coding Practice



Learning Objectives

By the end of this lesson, students will be able to:

- Understand what coding practice means.

FOUNDATION

- Read and understand simple programming problems.
- Break a problem into smaller steps.
- Write Python programs independently.
- Use variables, input, operators and conditions.
- Practice loops, lists and functions.
- Find and fix simple errors.
- Test programs with different inputs.
- Develop problem-solving skills.

1. What is Coding Practice?

Coding practice means writing programs regularly to improve programming skills.

Just like:

A cricket player practices batting.

A musician practices playing.

A student practices writing.

A programmer practices coding.

The more problems you solve, the better you become at programming.



3DWEBSOFT FOUNDATION

2. Why is Coding Practice Important?

Coding practice helps students learn how to:

- Think logically
- Solve problems
- Find mistakes
- Create solutions
- Write better programs
- Become confident programmers

3. The Coding Practice Cycle

Every programmer follows a simple process:

PROBLEM

↓

THINK

↓

PLAN

↓

CODE

↓

RUN

↓

TEST

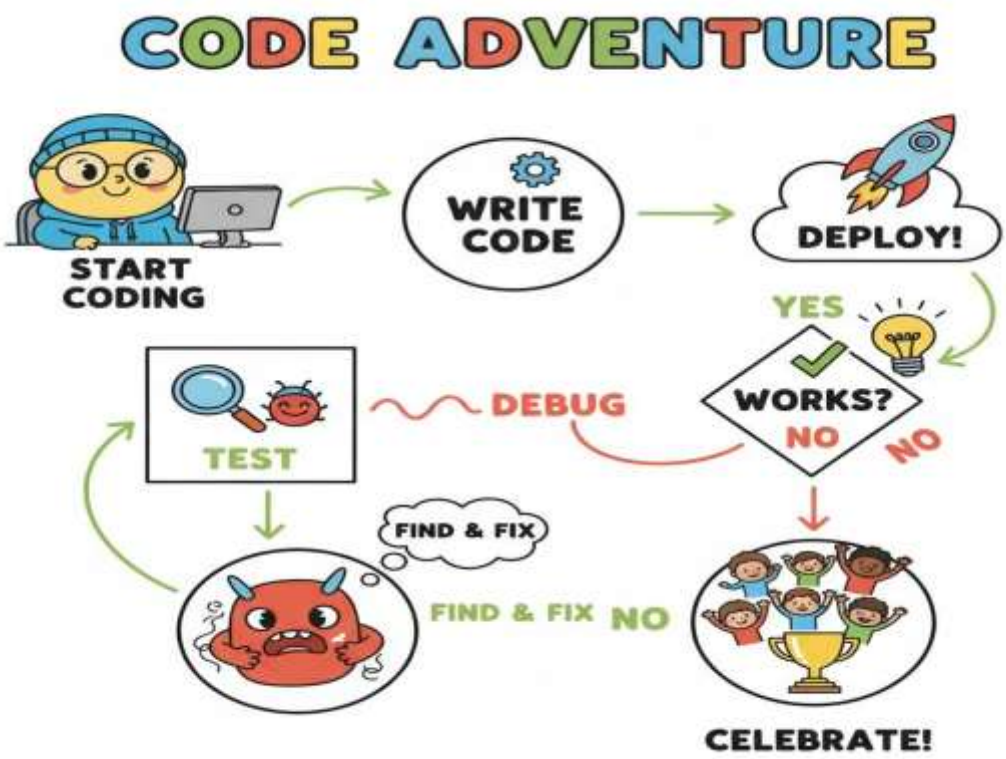
↓

FIX ERRORS

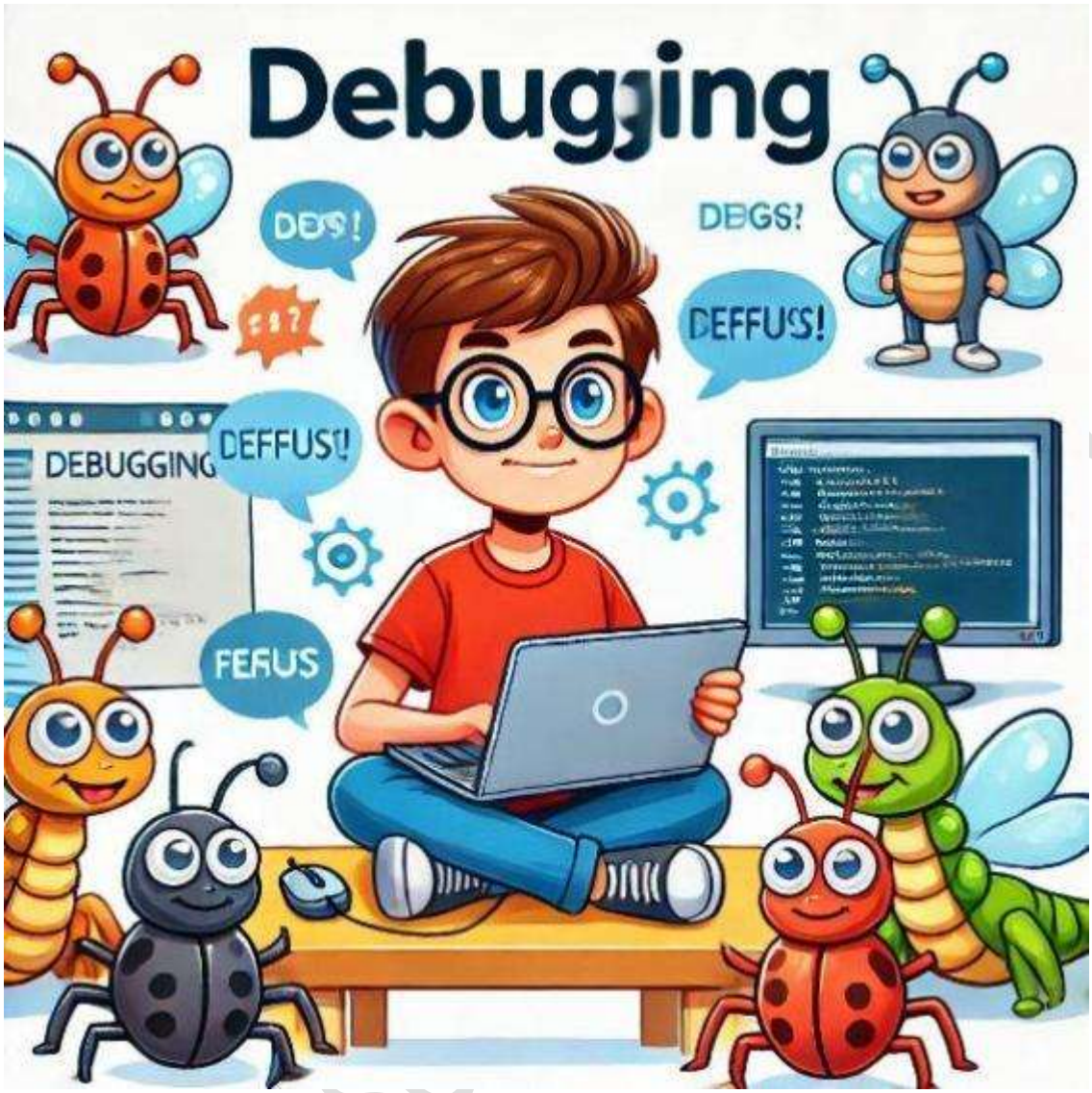
↓

3DWEBSOFT FOUNDATION

SOLVE


CODE ADVENTURE
START CODING
WRITE CODE
DEPLOY!
YES
WORKS?
NO
NO
DEBUG
TEST
FIND & FIX
FIND & FIX
NO
CELEBRATE!


Understand problem
Design the solution
Implement
Test and debug
Refactor and maintain



Step 1 – Understand the Problem

Before writing code, ask:

What do I need to do?

Example:

Write a program to find whether a number is even or odd.

We need:

Input: A number

Process: Check whether it is divisible by 2

Output: Even or Odd

Step 2 – Make a Plan

Before coding:

1. Get a number.

2. Divide it by 2.

3. Check the remainder.

4. If remainder is 0 → Even.

5. Otherwise → Odd.

Step 3 – Write the Code

```
number = int(input("Enter a number: "))


if number % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

Step 4 – Test the Code

Test 1

Input:

8

Output:

Even number

Test 2

3DWEBSOFT FOUNDATION

Input:

7

Output:

Odd number

Step 5 – Debug

Debugging means finding and fixing mistakes in a program.

Example:

```
number = int(input("Enter number: ")


print(number)
```

There is a missing).

Correct:

```
number = int(input("Enter number: "))


print(number)
```

Important Python Skills for Practice

Students should practice:

Skill	Example
Variables	age = 13
Input	input()
Output	print()
Operators	+ - * / %

3DVEBSOFT FOUNDATION

Skill	Example
Conditions	if
Loops	for, while
Lists	[1, 2, 3]
Functions	def
Exceptions	try/except

Practical Session

Practice 1 – Greeting Program

Problem

Ask the student's name and display a greeting.

```
name = input("Enter your name: ")


print("Hello", name)
print("Welcome to Python!")
```

Sample Output

```
Enter your name: Aisha
Hello Aisha
Welcome to Python!
```

Practice 2 – Add Two Numbers

Problem

Take two numbers and calculate their sum.

```
a = int(input("Enter first number: "))
```

3DWEBSOFT FOUNDATION

```
b = int(input("Enter second number: "))

total = a + b

print("Sum:", total)
```

Example

Enter first number: 25

Enter second number: 15

Sum: 40

Practice 3 – Calculate Area

Problem

Find the area of a rectangle.

Formula

Area = Length × Width

Code

```
length = float(input("Enter length: "))
width = float(input("Enter width: "))

area = length * width

print("Area:", area)
```

Practice 4 – Even or Odd

3DWEBSOFT FOUNDATION

```
number = int(input("Enter a number: "))


if number % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Challenge

Modify the program to display:

Even number

or:

Odd number

Practice 5 – Positive, Negative or Zero

Problem

Determine whether a number is:

- Positive
- Negative
- Zero

Code

```
number = int(input("Enter a number: "))


if number > 0:
    print("Positive")
elif number < 0:
    print("Negative")
```

3DWEBSOFT FOUNDATION

```
else:
    print("Zero")
```

Practice 6 – Find the Largest Number

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

if a > b:
    print("Largest:", a)
else:
    print("Largest:", b)
```

Challenge

Modify it to find the largest of **three numbers**.

Practice 7 – Multiplication Table

Problem

Print the multiplication table of a number.

```
number = int(input("Enter a number: "))

for i in range(1, 11):
    print(number, "×", i, "=", number * i)
```

Example

For 5:

5 × 1 = 5

5 × 2 = 10

3DWEBSOFT FOUNDATION

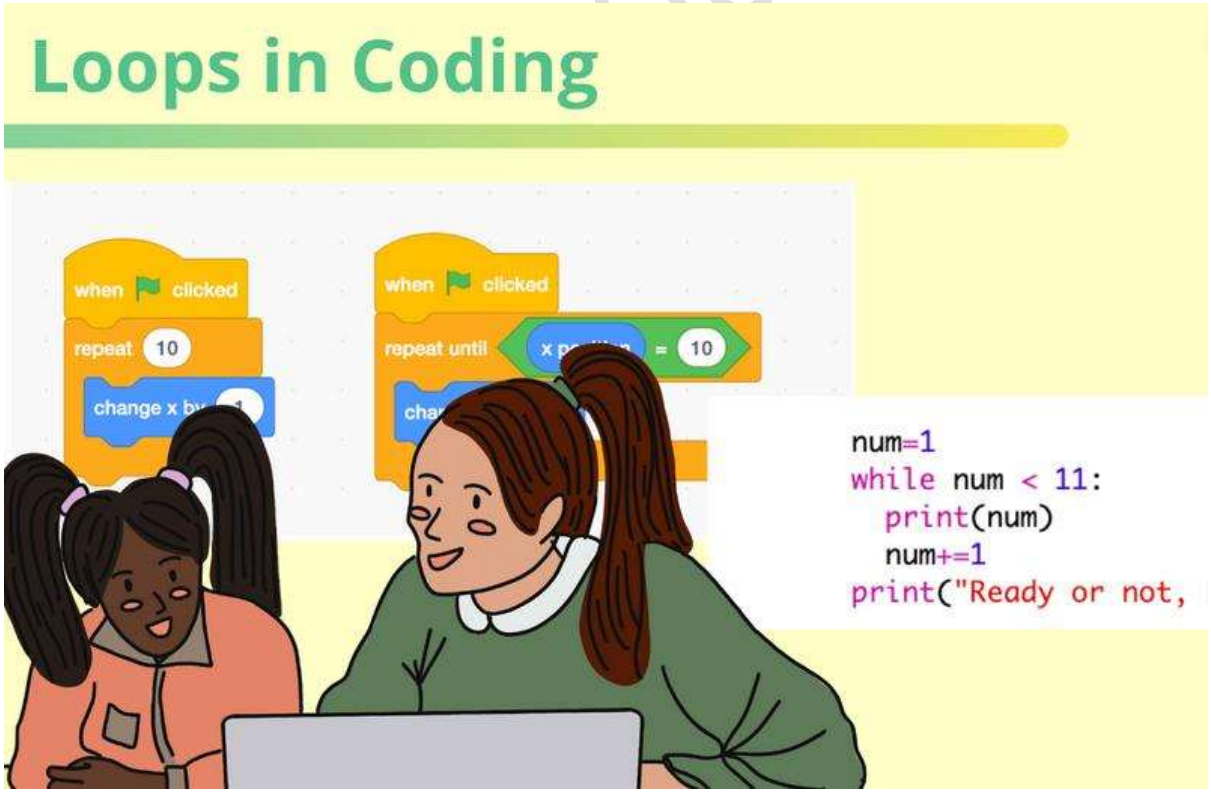
5 × 3 = 15

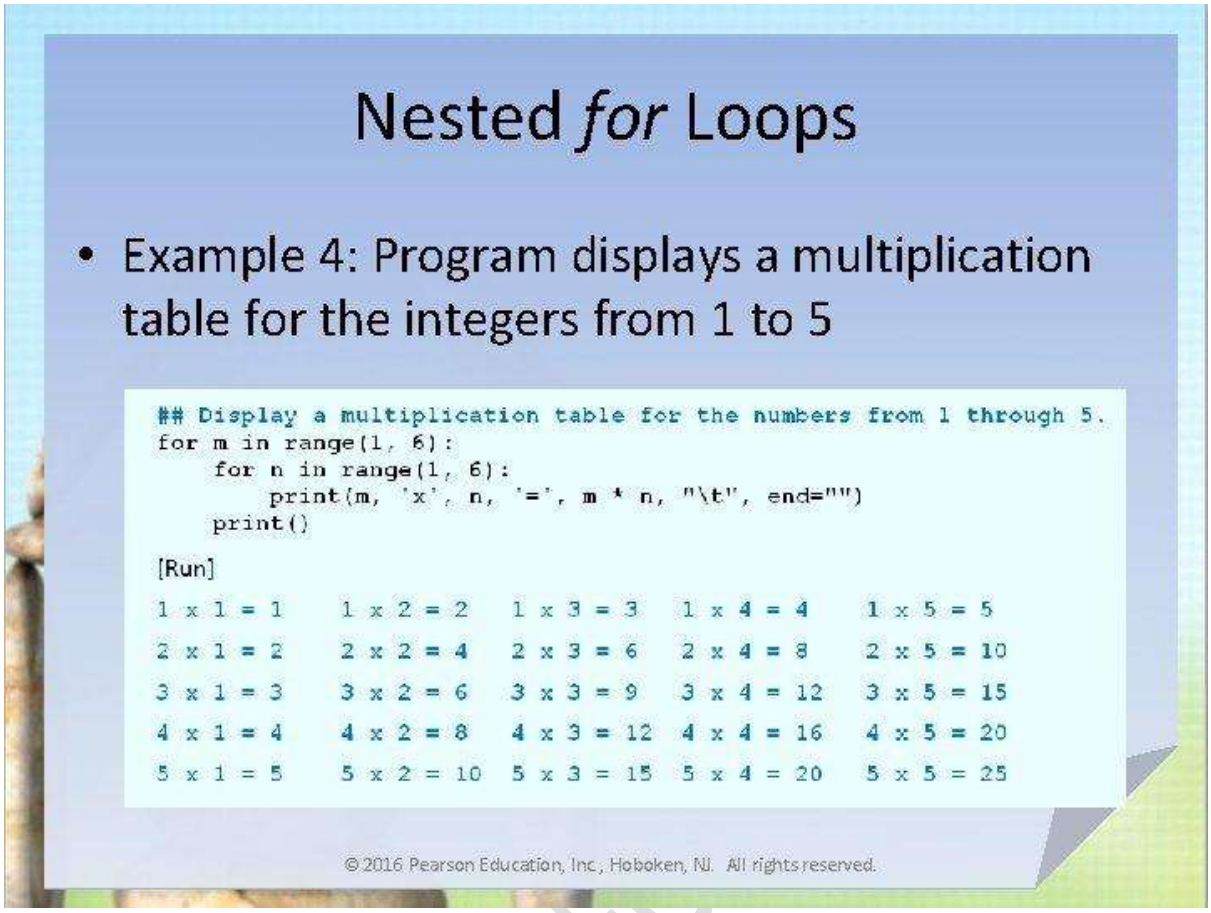
…

5 × 10 = 50

```
# Print table using for loop and range() function
for i in range(1,11):
    print("2 x " + str(i) + " = " + str(2 * i))
```

```
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20
```





Practice 8 – Count from 1 to 10

```
for i in range(1, 11):
    print(i)
```

Output:

1

2

3

4

5

6

7

3DWEBSOFT

8

9

10

Practice 9 – Sum from 1 to 10

```
total = 0


for i in range(1, 11):
    total = total + i


print("Total:", total)
```

Output:

Total: 55

Practice 10 – Countdown

```
for i in range(10, 0, -1):
    print(i)


print(" Go!")
```

Output:

10

9

8

7

6

3DWEBSOFT FOUNDATION

5

4

3

2

1

Go!

Practice 11 – List of Fruits

```
fruits = ["Apple", "Mango", "Banana", "Orange"]

for fruit in fruits:
    print(fruit)
```

Output:

Apple

Mango

Banana

Orange

Practice 12 – Add Numbers to a List

```
numbers = []

for i in range(5):
    number = int(input("Enter number: "))
    numbers.append(number)
```

3DWEBSOFT FOUNDATION

```
print("Numbers:", numbers)
```

Practice 13 – Find Total Marks

```
marks = [80, 75, 90, 85, 95]

total = sum(marks)

print("Total:", total)
```

Practice 14 – Find Highest Mark

```
marks = [80, 75, 90, 85, 95]

highest = max(marks)

print("Highest mark:", highest)
```

Practice 15 – Find Lowest Mark

```
marks = [80, 75, 90, 85, 95]

lowest = min(marks)

print("Lowest mark:", lowest)
```

Practice 16 – Count Characters

```
word = input("Enter a word: ")
```

3DWEBSOFT FOUNDATION

```
print("Number of characters:", len(word))
```

Example:

Enter a word: Python

Number of characters: 6

Practice 17 – Reverse a Word

```
word = input("Enter a word: ")

print("Reverse:", word[::-1])
```

Example:

Python

nohtyP

Practice 18 – Check a Word

```
word = input("Enter a word: ")

if word == "Python":
    print("Correct!")
else:
    print("Try again!")
```

Practice 19 – Number Guessing Game

```
secret = 7
```

3DWEBSOFT FOUNDATION

```
guess = int(input("Guess the number: "))


if guess == secret:
    print(" Correct!")
elif guess < secret:
    print("Too low!")
else:
    print("Too high!")
```

Challenge

Allow the player **3 attempts** using a loop.

Practice 20 – Simple Calculator

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))


operator = input("Enter +, -, *, or /: ")


if operator == "+":
    print("Answer:", a + b)


elif operator == "-":
    print("Answer:", a - b)


elif operator == "*":
    print("Answer:", a * b)
```

3DWEBSOFT FOUNDATION

```
elif operator == "/":
    if b != 0:
        print("Answer:", a / b)
    else:
        print("Cannot divide by zero.")

else:
    print("Invalid operator.")
```

Practice 21 – Safe Input

```
try:
    number = int(input("Enter a number: "))
    print("You entered:", number)

except ValueError:
    print(" Please enter a number.")
```

Practice 22 – Safe Division

```
try:
    a = int(input("Enter first number: "))
    b = int(input("Enter second number: "))

    print("Answer:", a / b)
```

3DWEBSOFT FOUNDATION

```
except ValueError:
    print("Please enter numbers only.")

except ZeroDivisionError:
    print("Cannot divide by zero.")
```

Problem-Solving Challenge

Challenge 1 – Age Checker

Write a program that asks for age.

If:

age >= 18

display:

You are an adult.

Otherwise:

You are a minor.

Solution

```
age = int(input("Enter your age: "))

if age >= 18:
    print("You are an adult.")
else:
    print("You are a minor.")
```

Challenge 2 – Temperature Checker

Ask for temperature.

3DWEBSOFT FOUNDATION

Above 35 → Very Hot

25–35 → Warm

15–24 → Cool

Below 15 → Cold

Solution

```
temperature = float(input("Enter temperature: "))


if temperature > 35:
    print("Very Hot")
elif temperature >= 25:
    print("Warm")
elif temperature >= 15:
    print("Cool")
else:
    print("Cold")
```

Challenge 3 – Password Checker

```
password = input("Enter password: ")


if password == "python123":
    print("Access granted!")
else:
    print("Wrong password!")
```

Challenge 4 – Quiz Game

3DWEBSOFT FOUNDATION

```
score = 0

answer = input("What is the capital of India? ")

if answer.lower() == "new delhi":
    print("Correct!")
    score += 1
else:
    print("Wrong!")

print("Score:", score)
```

23. What is Debugging?

Debugging means finding and correcting errors in a program.

There are three common types of problems students should know:

1. Syntax Error

The Python grammar is wrong.

```
print("Hello"
```

Correct:

```
print("Hello")
```

2. Runtime Error

The program starts but encounters a problem while running.

Example:

```
print(10 / 0)
```

3DWEBSOFT FOUNDATION

3. Logic Error

The program runs but gives the wrong answer.

Example:

a = 10

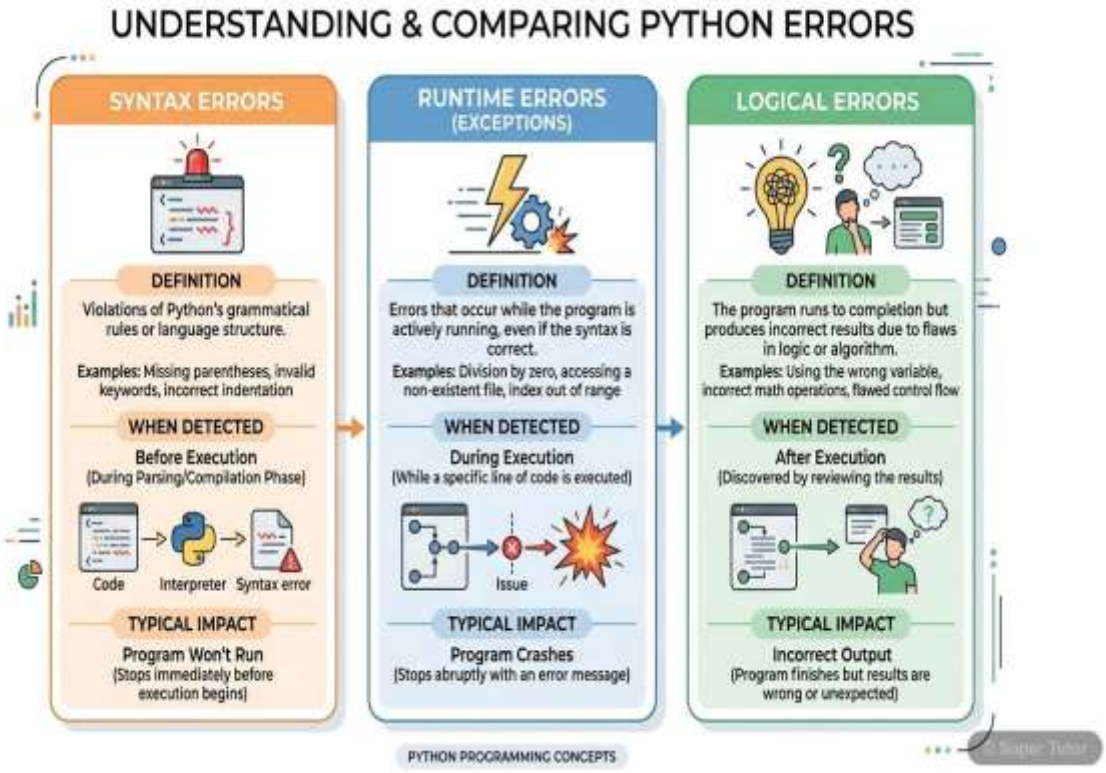
b = 5

print(a - b)

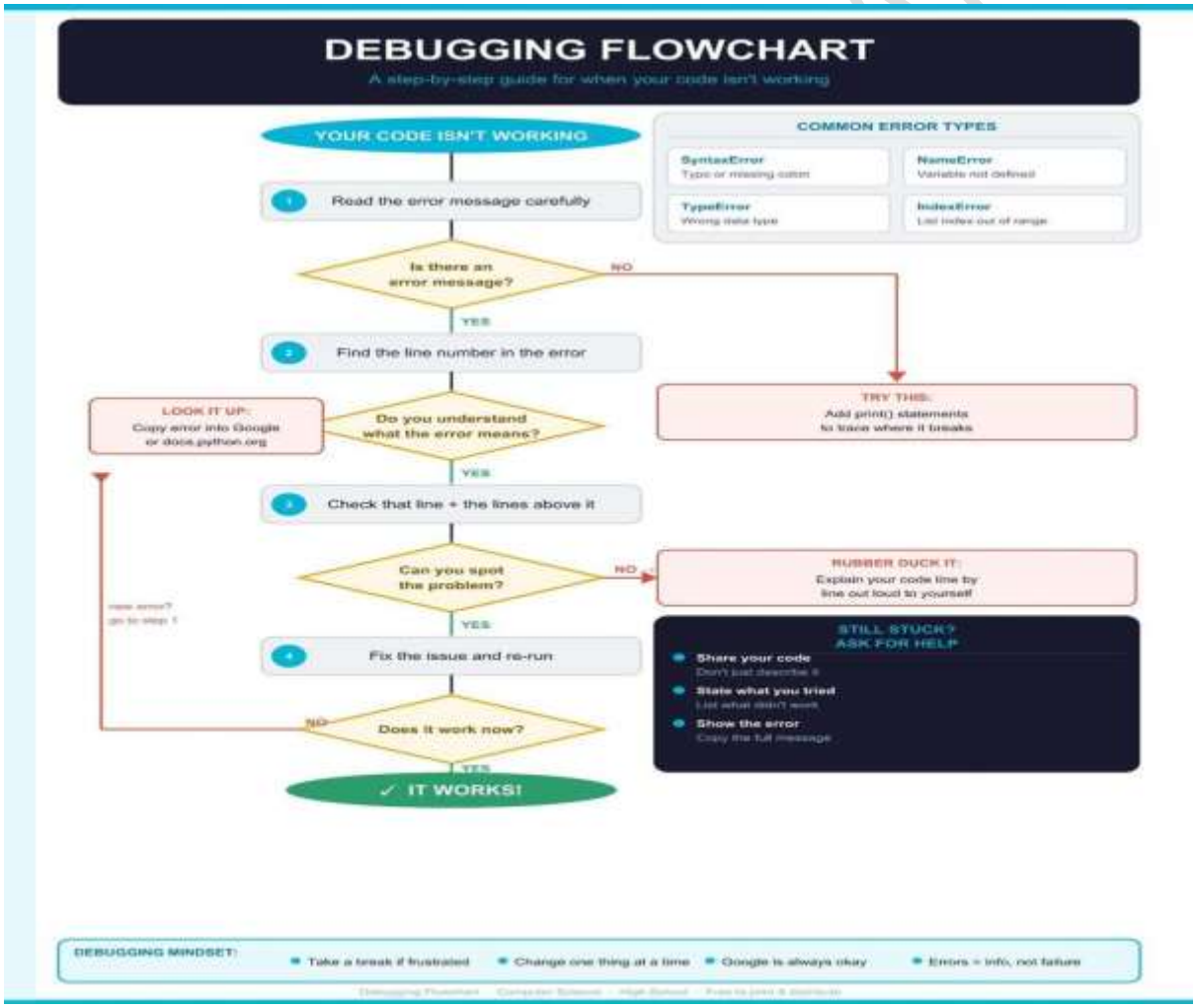
If we wanted the sum, the program runs successfully but gives the wrong result.

Correct:

print(a + b)







24. Debugging Steps

When your program doesn't work:

Step 1

Read the error message.

Step 2

Look at the line number.

Step 3

Check spelling.

Step 4

Check brackets.

Step 5

Check indentation.

Step 6

Test the program again.

Step 7

Ask:

"What did I expect?"

and

"What did I actually get?"

25. Debug This Code

Wrong Code

```
name = input("Enter name: ")


if name == "Aisha"
   print("Hello Aisha")
```

3DWEBSOFT FOUNDATION

Find the Error

There is a missing :.

Correct Code

```
name = input("Enter name: ")


if name == "Aisha":
    print("Hello Aisha")
```

26. Debug This Code

Wrong

```
for i in range(5)
    print(i)
```

Correct

```
for i in range(5):
    print(i)
```

27. Coding Practice Rules

Students should follow these rules:

Rule 1

Understand before coding.

Rule 2

Start with small steps.

Rule 3

Test your code frequently.

Rule 4

3DWEBSOFT FOUNDATION

Don't be afraid of errors.

Rule 5

Read error messages carefully.

Rule 6

Try different inputs.

Rule 7

Explain your own code.

28. Mini Coding Challenge Set

Beginner Level

1. Print your name.
2. Add two numbers.
3. Find the area of a rectangle.
4. Check even or odd.
5. Check positive or negative.
6. Find the largest of two numbers.
7. Print numbers 1–20.
8. Print a multiplication table.

Intermediate Level

9. Find the largest of three numbers.
10. Find the sum of a list.
11. Find the highest mark.
12. Find the lowest mark.
13. Count vowels.

3DWEBSOFT FOUNDATION

14. Reverse a word.
15. Create a simple calculator.
16. Create a guessing game.

Challenge Level

17. Quiz game.
18. Student grade calculator.
19. To-do list.
20. Number guessing game with 3 attempts.
21. Simple shopping list.
22. Student report card.
23. Temperature converter.
24. Simple password checker.

29. Final Coding Challenge

Student Performance Analyzer

Create a program that:

Input

Ask for:

- Student name
- 5 subject marks

Process

Calculate:

- Total
- Average

3DWEBSOFT FOUNDATION

- Highest mark
- Lowest mark
- Grade

Output

Display a beautiful report.

Starter Code

```
name = input("Enter student name: ")


marks = []


for i in range(5):
    mark = int(input("Enter mark: "))
    marks.append(mark)


total = sum(marks)
average = total / len(marks)
highest = max(marks)
lowest = min(marks)


if average >= 90:
    grade = "A+"
elif average >= 80:
    grade = "A"
elif average >= 70:
    grade = "B"
```

3DWEBSOFT FOUNDATION

```
elif average >= 60:
    grade = "C"
elif average >= 40:
    grade = "D"
else:
    grade = "F"

print()
print("===== STUDENT PERFORMANCE =====")
print("Name:", name)
print("Marks:", marks)
print("Total:", total)
print("Average:", round(average, 2))
print("Highest:", highest)
print("Lowest:", lowest)
print("Grade:", grade)
```

30. Coding Competition Activity

Divide the class into teams.

Team 1

Calculator

Team 2

Quiz Game

Team 3

Number Guessing Game

3DWEBSOFT FOUNDATION

Team 4

Student Performance Analyzer

Give each team **30 minutes**.

Then each team gets **3 minutes** to demonstrate the program.

31. Student Coding Worksheet

No.	Problem	Completed
1	Greeting Program	☐
2	Add Two Numbers	☐
3	Even/Odd	☐
4	Positive/Negative	☐
5	Largest Number	☐
6	Multiplication Table	☐
7	List Practice	☐
8	Marks Calculator	☐
9	Calculator	☐
10	Guessing Game	☐
11	Quiz Game	☐
12	Final Challenge	☐

3DWEBSOFT FOUNDATION

Quiz

1. What is coding practice?

A) Watching videos
B) Regularly solving programming problems
C) Playing games
D) Drawing pictures

Answer: B

2. What should we do before coding?

A) Guess
B) Understand the problem
C) Close Python
D) Delete the program

Answer: B

3. Which symbol checks the remainder?

A) +

B) %

C) *

D) #

Answer: B

4. Which keyword is used for a condition?

A) if

B) check

C) when

D) test

Answer: A

5. Which keyword creates a loop?

A) repeat

B) for

C) again

3DWEBSOFT FOUNDATION

D) loop

Answer: B

6. What does debugging mean?

A) Writing a new computer
B) Finding and fixing errors
C) Deleting Python
D) Printing code

Answer: B

7. Which function finds the largest value?

A) high()

B) max()

C) largest()

D) top()

Answer: B

8. Which function finds the total of a list?

A) sum()

B) total()

C) add()

D) plus()

Answer: A

3DWEBSOFT FOUNDATION

Homework

Complete these programs:

Task 1

Create an **Even/Odd Checker**.

Task 2

Create a **Multiplication Table Generator**.

Task 3

Create a **Marks Calculator**.

Task 4

Create a **Number Guessing Game**.

Task 5

Create a **Quiz Game with 5 Questions**.

34. Learning Check

Students should now be able to:

☑ Understand a programming problem.
☑ Plan a solution.
☑ Write Python code.
☑ Test different inputs.
☑ Find simple errors.
☑ Debug programs.
☑ Use loops and lists.
☑ Use conditions.
☑ Use functions.
☑ Handle exceptions.
☑ Build small projects independently.

3DWEBSOFT FOUNDATION

One-Minute Revision

PROBLEM

↓

THINK

↓

PLAN

↓

CODE

↓

RUN

↓

TEST

↓

DEBUG

↓

SOLVE

3DWEBSOFT FOUNDATION